

Geometry  
Unit 12  
Lesson 9 Homework

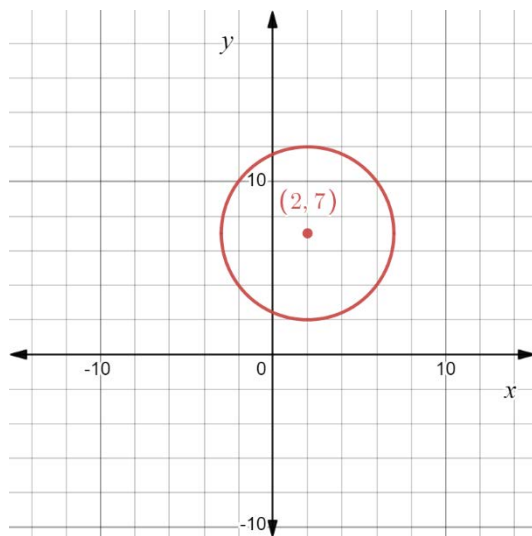
Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

For questions 1 and 2, consider that the equation of circle  $S$  is  $(x - 2)^2 + (y - 7)^2 = 25$ .

1. Give the key features of the circle.

| Center             | Radius             |
|--------------------|--------------------|
| $(2, 7)$           | 5                  |
| Domain             | Range              |
| $-3 \leq x \leq 7$ | $2 \leq y \leq 12$ |

2. Graph circle  $S$ .

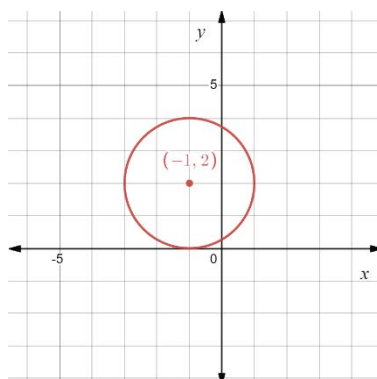


For questions 3 and 4, consider that the equation of circle  $K$  is  $(x + 1)^2 + (y - 2)^2 = 4$ .

3. Give the key features of the circle.

| Center             | Radius            |
|--------------------|-------------------|
| $(-1, 2)$          | 2                 |
| Domain             | Range             |
| $-3 \leq x \leq 1$ | $0 \leq y \leq 4$ |

4. Graph circle  $K$ .



For questions 5 through 7, consider that the equation of circle *L* is  $x^2 + 2x + y^2 + 2y = 7$ .

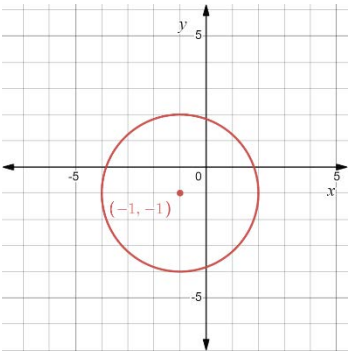
5. Rewrite the equation of the circle in center-radius form.

$$\begin{aligned}x^2 + 2x + 1 + y^2 + 2y + 1 &= 7 + 1 + 1 \\(x + 1)^2 + (y + 1)^2 &= 9\end{aligned}$$

6. Give the key features of the circle.

| Center             | Radius             |
|--------------------|--------------------|
| $(-1, -1)$         | 3                  |
| Domain             | Range              |
| $-4 \leq x \leq 2$ | $-4 \leq y \leq 2$ |

7. Graph circle .



For questions 8 through 10, consider that the equation of circle *Z* is  $x^2 - 6x + y^2 + 4y - 12 = 0$ .

8. Rewrite the equation of the circle in center-radius form.

$$\begin{aligned}x^2 - 6x + 9 + y^2 + 4y + 4 &= 12 + 9 + 4 \\(x - 3)^2 + (y + 2)^2 &= 25\end{aligned}$$

9. Give the key features of the circle.

| Center             | Radius             |
|--------------------|--------------------|
| $(3, -2)$          | 5                  |
| Domain             | Range              |
| $-2 \leq x \leq 8$ | $-7 \leq y \leq 3$ |

10. Graph circle *Z*.

